

Solutions of Problems on CH 27

1. A 200-km-long high-voltage transmission line 2.00 cm in diameter carries a steady current of 1 000 A. If the conductor is copper with a free charge density of 8.50×10^{28} electrons per cubic meter, how many years does it take one electron to travel the full length of the cable?

$$v_d = \frac{I}{nqa} = \frac{I}{n|e|(\pi d^2 / 4)} \Rightarrow \Delta t = \frac{L}{v_d} = \frac{Ln|e|(\pi d^2)}{4I}$$

$$\Delta t = \frac{(200 \times 10^3 \text{ m})(8.50 \times 10^{28} \text{ m}^{-3})(1.60 \times 10^{-19} \text{ C})\pi(0.02 \text{ m})^2}{4(1\,000 \text{ A})}$$

$$= (8.55 \times 10^8 \text{ s}) \left(\frac{1 \text{ yr}}{3.156 \times 10^7 \text{ s}} \right) = \boxed{27.1 \text{ yr}}$$



5. A proton beam in an accelerator carries a current of $125\ \mu\text{A}$. If the beam is incident on a target, how many protons strike the target in a period of $23.0\ \text{s}$?

$$N = \frac{I(\Delta t)}{e} = \frac{(125 \times 10^{-6}\ \text{C/s})(23.0\ \text{s})}{1.60 \times 10^{-19}\ \text{C/proton}} = \boxed{1.80 \times 10^{16}\ \text{protons}}$$

9. The quantity of charge q (in coulombs) that has passed **W** through a surface of area 2.00 cm^2 varies with time according to the equation $q = 4t^3 + 5t + 6$, where t is in seconds. (a) What is the instantaneous current through the surface at $t = 1.00 \text{ s}$? (b) What is the value of the current density?

We are given $q = 4t^3 + 5t + 6$. The area is

$$A = (2.00 \text{ cm}^2) \left(\frac{1.00 \text{ m}}{100 \text{ cm}} \right)^2 = 2.00 \times 10^{-4} \text{ m}^2$$

$$(a) \quad I(1.00 \text{ s}) = \left. \frac{dq}{dt} \right|_{t=1.00 \text{ s}} = (12t^2 + 5) \Big|_{t=1.00 \text{ s}} = \boxed{17.0 \text{ A}}$$

$$(b) \quad J = \frac{I}{A} = \frac{17.0 \text{ A}}{2.00 \times 10^{-4} \text{ m}^2} = \boxed{85.0 \text{ kA/m}^2}$$

15. A wire 50.0 m long and 2.00 mm in diameter is connected to a source with a potential difference of 9.11 V, and the current is found to be 36.0 A. Assume a temperature of 20.0°C and, using Table 27.2, identify the metal out of which the wire is made.

$$R = \rho \ell / A = \rho \ell / (\pi d^2 / 4)$$
$$\rho = \left(\frac{\pi d^2}{4 \ell} \right) R = \left(\frac{\pi d^2}{4 \ell} \right) \left(\frac{\Delta V}{I} \right) = \left[\frac{\pi (2.00 \times 10^{-3} \text{ m})^2}{4 (50.0 \text{ m})} \right] \left(\frac{9.11 \text{ V}}{36.0 \text{ A}} \right)$$
$$= 1.59 \times 10^{-8} \text{ } \Omega \cdot \text{m}$$

Then, from Table 27.2, we see that the wire is made of silver.

16. A 0.900-V potential difference is maintained across a 1.50-m length of tungsten wire that has a cross-sectional area of 0.600 mm². What is the current in the wire?

$$\Delta V = IR \quad \text{and} \quad R = \frac{\rho \ell}{A}.$$

$$A = (0.600 \text{ mm}^2) \left(\frac{1.00 \text{ m}}{1\,000 \text{ mm}} \right)^2 = 6.00 \times 10^{-7} \text{ m}^2$$

$$\Delta V = \frac{I \rho \ell}{A} \quad \rightarrow \quad I = \frac{\Delta V A}{\rho \ell} = \frac{(0.900 \text{ V})(6.00 \times 10^{-7} \text{ m}^2)}{(5.60 \times 10^{-8} \, \Omega \cdot \text{m})(1.50 \text{ m})}$$

$$I = \boxed{6.43 \text{ A}}$$

23. A current density of $6.00 \times 10^{-13} \text{ A/m}^2$ exists in the atmosphere at a location where the electric field is 100 V/m . Calculate the electrical conductivity of the Earth's atmosphere in this region.

$$J = \sigma E \quad \text{so} \quad \sigma = \frac{J}{E} = \frac{6.00 \times 10^{-13} \text{ A/m}^2}{100 \text{ V/m}} = \boxed{6.00 \times 10^{-15} (\Omega \cdot \text{m})^{-1}}.$$

26. A certain lightbulb has a tungsten filament with a resistance of $19.0\ \Omega$ when at 20.0°C and $140\ \Omega$ when hot. Assume the resistivity of tungsten varies linearly with temperature even over the large temperature range involved here. Find the temperature of the hot filament.

$$R = R_0 [1 + \alpha(\Delta T)]$$

$$140\ \Omega = (19.0\ \Omega) [1 + (4.50 \times 10^{-3}/^\circ\text{C}) \Delta T]$$

$$\Delta T = 1.42 \times 10^3\ ^\circ\text{C} = T - 20.0^\circ\text{C}$$

$$\boxed{T = 1.44 \times 10^3\ ^\circ\text{C}}$$

31. (a) A 34.5-m length of copper wire at 20.0°C has a radius of 0.25 mm. If a potential difference of 9.00 V is applied across the length of the wire, determine the current in the wire. (b) If the wire is heated to 30.0°C while the 9.00-V potential difference is maintained, what is the resulting current in the wire?

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(a) The resistance at 20.0°C is

$$R_0 = \frac{\rho \ell}{A} = \frac{(1.7 \times 10^{-8} \, \Omega \cdot \text{m})(34.5 \, \text{m})}{\pi (0.25 \times 10^{-3} \, \text{m})^2} = 2.99 \, \Omega$$

$$I = \frac{\Delta V}{R_0} = \frac{9.00 \, \text{V}}{3.00 \, \Omega} = \boxed{3.01 \, \text{A}}$$

(b) At 30.0°C , $R = R_0 [1 + \alpha(\Delta T)]$

$$= (2.99 \, \Omega) \left[1 + (3.9 \times 10^{-3} \, (^{\circ}\text{C})^{-1})(30.0^{\circ}\text{C} - 20.0^{\circ}\text{C}) \right] = 3.10 \, \Omega$$

$$I = \frac{\Delta V}{R_0} = \frac{9.00 \, \text{V}}{3.10 \, \Omega} = \boxed{2.90 \, \text{A}}$$

33. An aluminum wire with a diameter of 0.100 mm has a uniform electric field of 0.200 V/m imposed along its entire length. The temperature of the wire is 50.0°C. Assume one free electron per atom. (a) Use the information in Table 27.2 to determine the resistivity of aluminum at this temperature. (b) What is the current density in the wire? (c) What is the total current in the wire? (d) What is the drift speed of the conduction electrons? (e) What potential difference must exist between the ends of a 2.00-m length of the wire to produce the stated electric field?

$$(a) \quad \rho = \rho_0 [1 + \alpha(T - T_0)]$$

$$\begin{aligned} \rho &= (2.82 \times 10^{-8} \, \Omega \cdot \text{m}) [1 + (3.90 \times 10^{-3} \, ^\circ\text{C}^{-1})(30.0^\circ\text{C})] \\ &= \boxed{3.15 \times 10^{-8} \, \Omega \cdot \text{m}} \end{aligned}$$

$$(b) \quad J = \sigma E = \frac{E}{\rho} = \left(\frac{0.200 \, \text{V/m}}{3.15 \times 10^{-8} \, \Omega \cdot \text{m}} \right) \left(\frac{1 \, \Omega \cdot \text{A}}{\text{V}} \right) = \boxed{6.35 \times 10^6 \, \text{A/m}^2}$$

$$(c) \quad J = \frac{I}{A} = \frac{I}{\pi r^2}$$

$$I = J(\pi r^2) = (6.35 \times 10^6 \, \text{A/m}^2) [\pi (5.00 \times 10^{-5} \, \text{m})^2] = \boxed{49.9 \, \text{mA}}$$

$$\begin{aligned}
 \text{(d) } n &= \left(\frac{2.70 \times 10^3 \text{ kg}}{\text{m}^3} \right) \left(\frac{1 \text{ mol}}{26.98 \text{ g}} \right) \left(\frac{10^3 \text{ g}}{\text{kg}} \right) \left(\frac{6.02 \times 10^{23} \text{ free e}^-}{1 \text{ mol}} \right) \\
 &= 6.02 \times 10^{28} \text{ e}^- / \text{m}^3
 \end{aligned}$$

$$J = nqv_d$$

$$\begin{aligned}
 v_d &= \frac{J}{nq} = \frac{6.35 \times 10^6 \text{ A/m}^2}{(6.02 \times 10^{28} \text{ e}^- / \text{m}^3)(-1.60 \times 10^{-19} \text{ C/e}^-)} \\
 &= \boxed{-6.59 \times 10^{-4} \text{ m/s}}
 \end{aligned}$$

$$\text{(e) } \Delta V = E\ell = (0.200 \text{ V/m})(2.00 \text{ m}) = \boxed{0.400 \text{ V}}$$

37. In a hydroelectric installation, a turbine delivers 1 500 hp to a generator, which in turn transfers 80.0% of the mechanical energy out by electrical transmission. Under these conditions, what current does the generator deliver at a terminal potential difference of 2 000 V?

$$P = 0.800(1\,500\text{ hp})(746\text{ W/hp}) = 8.95 \times 10^5\text{ W}$$

Then, from $P = I\Delta V$,

$$I = \frac{P}{\Delta V} = \frac{8.95 \times 10^5\text{ W}}{2\,000\text{ V}} = \boxed{448\text{ A}}$$

39. A certain waffle iron is rated at 1.00 kW when connected to a 120-V source. (a) What current does the waffle iron carry? (b) What is its resistance?

$$(a) P = I\Delta V \rightarrow I = P/\Delta V = (1.00 \times 10^3 \text{ W})/(120 \text{ V}) = \boxed{8.33 \text{ A}}$$

$$(b) P = \Delta V^2/R \rightarrow R = \Delta V^2/P = (120 \text{ V})^2/(1.00 \times 10^3 \text{ W}) = \boxed{14.4 \text{ }\Omega}$$

49. A coil of Nichrome wire is 25.0 m long. The wire has a diameter of 0.400 mm and is at 20.0°C. If it carries a current of 0.500 A, what are (a) the magnitude of the electric field in the wire and (b) the power delivered to it? (c) **What If?** If the temperature is increased to 340°C and the potential difference across the wire remains constant, what is the power delivered?



$$R = \frac{\rho \ell}{A} = \frac{(1.50 \times 10^{-6} \, \Omega \cdot \text{m}) 25.0 \, \text{m}}{\pi (0.200 \times 10^{-3} \, \text{m})^2} = 298 \, \Omega$$

$$\Delta V = IR = (0.500 \, \text{A})(298 \, \Omega) = 149 \, \text{V}$$

$$(a) \quad E = \frac{\Delta V}{\ell} = \frac{149 \, \text{V}}{25.0 \, \text{m}} = \boxed{5.97 \, \text{V/m}}$$

$$(b) \quad P = (\Delta V)I = (149 \, \text{V})(0.500 \, \text{A}) = \boxed{74.6 \, \text{W}}$$

$$(c) \quad R = R_0 [1 + \alpha(T - T_0)] = (298 \, \Omega) [1 + (0.400 \times 10^{-3} / ^\circ\text{C}) 320^\circ\text{C}] \\ = 337 \, \Omega$$

$$I = \frac{\Delta V}{R} = \frac{149 \, \text{V}}{337 \, \Omega} = 0.443 \, \text{A}$$

$$P = (\Delta V)I = (149 \, \text{V})(0.443 \, \text{A}) = \boxed{66.1 \, \text{W}}$$